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Studierichting: Informatica
Oefeningenreeks 4A nr. 14.5

$$\int \frac{\ln x}{\sqrt{x}} dx$$

$$\begin{cases} u = \ln x \\ dv = \dfrac{1}{\sqrt{x}} dx \end{cases} \Rightarrow \begin{cases} du = \dfrac{1}{x} \\ v = 2\sqrt{x} \end{cases}$$

$$= 2\sqrt{x}\ln x - \int \frac{2\sqrt{x}}{x} dx$$

$$= 2\sqrt{x}\ln x - 2\int x^{-\frac{1}{2}} dx$$

$$= 2\sqrt{x}\ln x - \frac{2x^{\frac{1}{2}}}{\frac{1}{2}} + C$$

$$= 2\sqrt{x}\ln x - 4\sqrt{x} + C$$

References